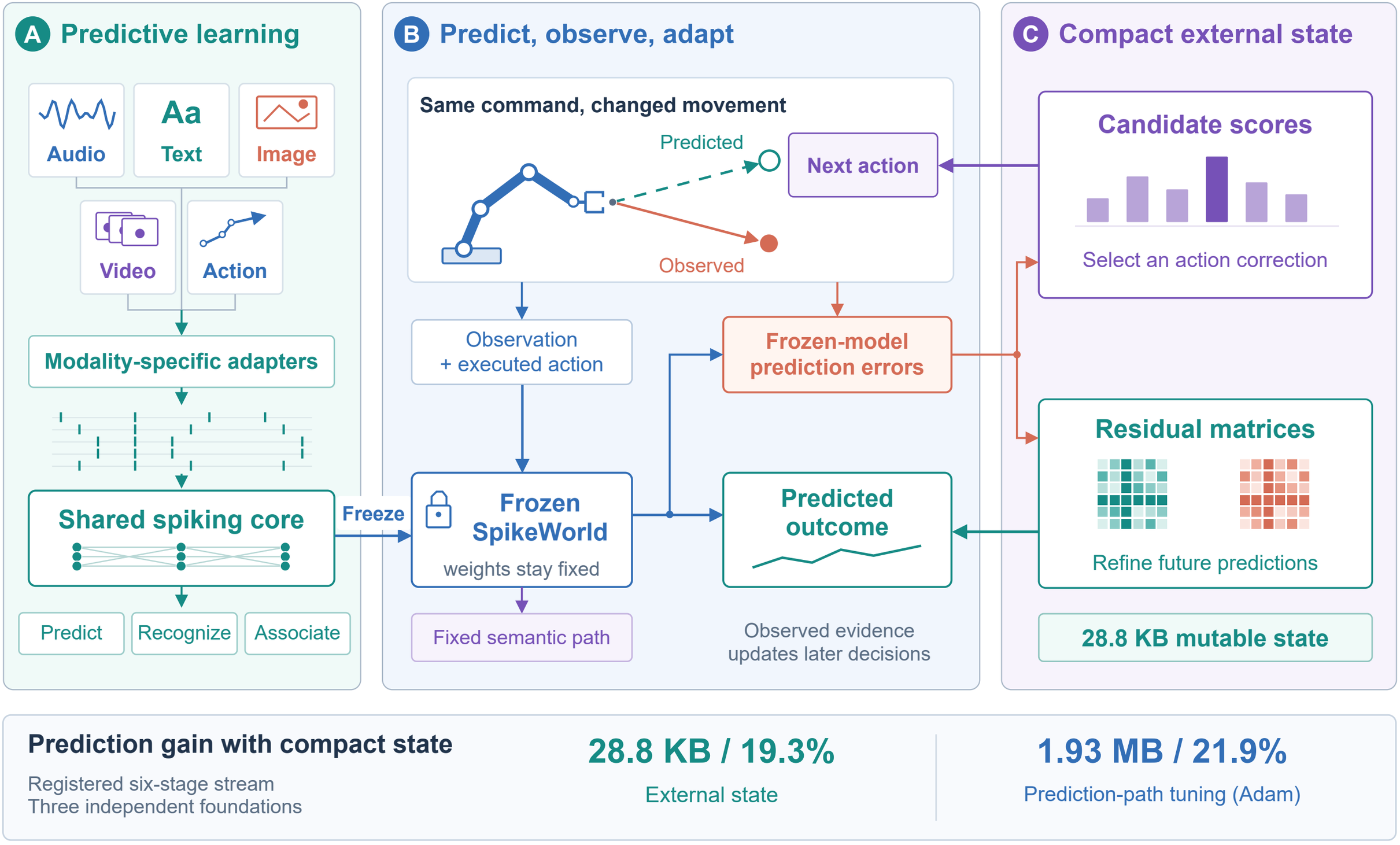

A Predictive learning
Audio
Text
Image
Video
Action
Modality-specific adapters
Shared spiking core
Predict
Recognize
Associate
Freeze
B Predict, observe, adapt
Same command, changed movement
Predicted
Observed
Next action
Observation
+ executed action
Frozen-model
prediction errors
Frozen
SpikeWorld
weights stay fixed
Predicted
outcome
Fixed semantic path
Observed evidence
updates later decisions
C Compact external state
Candidate scores
Select an action correction
Residual matrices
Refine future predictions
28.8 KB mutable state
Prediction gain with compact state
Registered six-stage stream
Three independent foundations
28.8 KB / 19.3%
External state
1.93 MB / 21.9%
Prediction-path tuning (Adam)